PROJECT BRIEF

AI Portfolio Research Dashboard

A unified, AI-assisted platform that turns fragmented investment research into a single decision-support workflow.

Role: Sole Designer, Developer & Architect   Duration: Mar–Apr 2026   Status: Completed

Live App: ai-portfolio-research.streamlit.app   Repository: github.com/venkatsaichand/AI_Portfolio_Research

Project at a Glance

7 Core Features	6 Dashboard Views	2 APIs Integrated
5 Portfolio Metrics Tracked	1 ML Forecasting Model	2 mo Build Duration

Executive Overview

Business Problem

Retail investors, finance students, and research analysts typically rely on separate platforms for pricing, benchmarking, forecasting, and interpretation. Each handoff between tools adds manual effort and breaks the analytical train of thought, making it harder to reach a confident, well-supported investment view.

Solution

The AI Portfolio Research Dashboard consolidates portfolio construction, performance tracking, benchmark comparison, forecasting, and AI-generated commentary into one interactive workspace — replacing a chain of disconnected tools with a single, continuous research session.

Business Value

- Eliminates spreadsheet-based portfolio tracking and manual data entry
- Streamlines research by replacing multiple tools with one workflow
- Improves decision quality through benchmark-aware performance context
- Reduces manual interpretation with AI-assisted, plain-language insights
- Keeps analysis current with real-time market data and no manual refresh

Core Features

Feature	Business Value
Portfolio Construction	Eliminates spreadsheet-based portfolio tracking by letting users build allocations directly in the app.
Interactive Portfolio Analytics	Enables faster interpretation of portfolio behavior through live charts and summary metrics.
Benchmark Comparison	Provides context for evaluating decisions against a market index, rather than viewing returns in isolation.
AI-Powered Investment Insights	Reduces manual interpretation of financial data via AI-generated, plain-language commentary.
Stock Price Forecasting	Supports scenario exploration and data-informed discussion with forward-looking projections.
Portfolio Optimization Suggestions	Guides allocation decisions using risk-return trade-off analysis.
Real-Time Financial Data	Keeps analysis current without manual updates via live market data retrieval.

Key Differentiators

Most portfolio dashboards stop at charting prices. This platform connects the full research sequence — construction, benchmarking, AI interpretation, forecasting, and optimization — into a single workflow, reflecting how research actually happens rather than how a chart library happens to display data.

Real-World Applications

- Personal investment tracking and portfolio review
- Financial education in universities and finance clubs
- Early-stage research support for wealth management and equity research teams
- Illustrative use in business analytics and consulting case discussions

Architecture & Engineering

User → Frontend → Data Source → Processing → AI / ML → Visualization → Insights

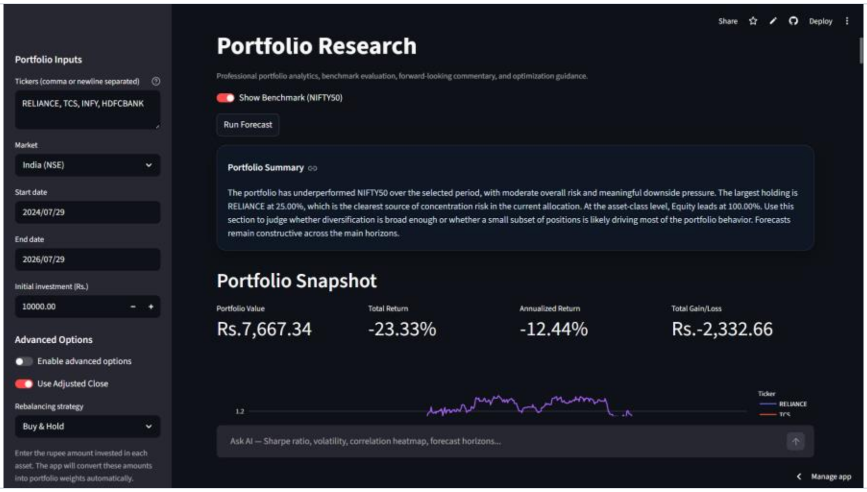
Key Engineering Decisions

Decision	Why
Streamlit	Delivers an interactive dashboard from a single Python codebase, prioritizing iteration speed on analytics over custom front-end tooling — appropriate for a research tool, not a consumer product.
Yahoo Finance API	Offers broad, dependable coverage of NSE-listed equities without licensing overhead, suiting a research context rather than a trading system with strict latency guarantees.
Google Gemini API	Used to translate structured portfolio metrics into grounded, plain-language commentary — an interpretation layer, not a general-purpose chatbot.
Random Forest Regressor	Captures non-linear price patterns without requiring large training datasets or heavy tuning, a practical trade-off over deep learning for single-portfolio forecasting.
Modern Portfolio Theory	An established, explainable framework for allocation trade-offs — important when suggestions are shown to non-technical users who need to trust the reasoning.
Plotly	Chosen over static plotting for zoom and hover interactivity, which supports exploratory analysis rather than one-off reporting.

Solution Architecture

- Portfolio analytics engine covering return, volatility, Sharpe ratio, and drawdown
- Correlation and risk-concentration analysis across holdings
- Prompt-engineered AI layer grounded in the portfolio's own computed metrics
- Deployed on Streamlit Community Cloud for public accessibility

Execution & Outlook



Portfolio Research overview — consolidated performance summary, benchmark toggle, and AI query bar in one view.

Challenges Solved

Combining live market data, a machine learning model, and a generative AI layer inside one responsive interface required reconciling data frequencies, normalizing assets for fair comparison, and keeping AI commentary grounded in actual computed metrics rather than generic output — the core data engineering and product-trust problems behind the platform.

Skills Demonstrated

Business & Analytical	Technical & Communication
Business analysis · Financial analysis · Product thinking · Decision support · Benchmark & forecast interpretation	Python development · API integration · Machine learning · Prompt engineering · Dashboard design · Data storytelling

Future Roadmap

- Deeper risk analytics, including position-level attribution
- News sentiment analysis layered into the AI insight engine
- Multi-user authentication for shared or client-facing use
- Portfolio alerts and a conversational AI investment assistant

Why This Project Matters

The AI Portfolio Research Dashboard was built to close a real gap in how investment research gets done — not to add another chart to an already crowded space. It reflects the ability to identify a workflow problem, scope a solution around it, and execute across data engineering, machine learning, applied AI, and product design to deliver something a real user would choose to keep using.